

## Q1 What is Artificial Intelligence?

Artificial intelligence (AI) is simply the ability of machines and computers to do things that normally need human smarts—like learning from experience, solving problems, making decisions, understanding language, and even creating new things.

Instead of just doing exactly what they're told step-by-step, AI systems can analyze data, spot patterns, and then use what they've learned to make smart choices or predictions. You've probably already used AI today without realizing it—like when Spotify suggests a song you end up loving, or when Gmail finishes writing your email for you.

## Q2 Difference between AI, Machine Learning, Generative AI, and Agentic AI.

| Technology                          | What It                                                                                                          | How It Works                                                                                     | Example                                                                               | Think of It As...                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Artificial Intelligence (AI)</b> | The broad field of creating machines that can mimic human intelligence.                                          | Uses rules, logic, and data to perform tasks that normally require human thinking.               | Siri, Google Maps, Chess-playing computers.                                           | <b>The entire universe of smart machines.</b>                            |
| <b>Machine Learning (ML)</b>        | A subset of AI that enables systems to learn from data without being explicitly programmed.                      | Finds patterns in data and improves its performance through experience.                          | Netflix recommendations, spam email filters.                                          | <b>A student that learns from examples.</b>                              |
| <b>Generative AI</b>                | A type of AI that can create new content such as text, images, music, or code.                                   | Learns patterns from large datasets and generates original outputs.                              | ChatGPT, DALL·E, Midjourney, Suno AI.                                                 | <b>A creative artist that produces new content.</b>                      |
| <b>Agentic AI</b>                   | An advanced form of AI that can make decisions, plan actions, and work toward goals with minimal human guidance. | Understands objectives, takes actions, evaluates results, and adjusts its approach autonomously. | AI assistants that can book meetings, research topics, and complete multi-step tasks. | <b>A personal assistant that can think, plan, and act independently.</b> |

## Q3. Explain any three open-source AI models available on Hugging Face.

### 1. BERT (Bidirectional Encoder Representations from Transformers)

BERT is like a smart reading assistant that understands context really well. It reads text from both directions (left-to-right and right-to-left) at the same time, which helps it grasp the full

meaning of words based on what comes before and after them. You can use it for tasks like answering questions, checking if a sentence is positive or negative, or even filling in missing words. It's lightweight, fast, and works great on regular computers, making it perfect for beginners.

## 2. Llama (by Meta)

Llama is a powerful language model that can chat, write stories, answer questions, and even help with coding. Think of it as a very knowledgeable assistant that's been trained on tons of text from the internet. The best part? Meta made it open-source, so anyone can use it for free. There are different sizes available—from small ones that run on laptops to huge ones for serious tasks. It's optimized for conversation and understands instructions really well.

## 3. Stable Diffusion

This is an image-generating model that creates pictures from text descriptions. You type something like "a sunset over mountains" and it paints that scene for you. What makes it special is that it's incredibly fast and produces high-quality images. Unlike some other image models that need expensive supercomputers, Stable Diffusion can run on a regular home computer with a decent graphics card. It's become the go-to choice for artists and creators who want AI-generated art.

Q4 What is the difference between traditional programming and AI-assisted development?

| Aspect                  | Traditional Programming                                                   | AI-Assisted Development                                                                                |
|-------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>How you work</b>     | You manually write all code, search documentation, and debug line-by-line | You describe your goal in natural language; AI suggests code you review and accept                     |
| <b>Who's in control</b> | You do everything yourself                                                | You stay in control—reviewing, validating, and making final decisions; AI is your helper               |
| <b>Speed</b>            | Slower; depends heavily on your expertise and time                        | Much faster; AI acts as a "productivity multiplier" that reduces the effort from idea to finished code |
| <b>Cognitive load</b>   | High—you remember everything, solve every problem manually                | Lower—AI handles repetitive tasks, so you focus on bigger picture decisions                            |
| <b>What changes</b>     | The code you write                                                        | The workflow: AI augments your skills, doesn't replace them                                            |

Q5 Explain AI hallucinations with one example

**AI hallucinations** are when an AI model generates false, misleading, or completely made-up information but presents it as if it's factually correct. The AI isn't "lying" on purpose—it's just predicting what words should come next based on patterns it learned during training, even when those predictions don't match reality.

## Example

### **The Fake Citation Problem:**

If you ask a chatbot like ChatGPT or Gemini to provide academic sources or peer-reviewed references for a topic, it might confidently invent completely fake papers, including:

- Nonexistent paper titles
- Made-up authors who never wrote anything
- Fake journal names
- Citations that don't exist anywhere